The Central Element of Algebraic R-Flux: A Malcev Algebra, Its Universal Envelope, and W as a Structural $\hbar$

Harald Rønning

Question. The algebraic R-flux bracket obtained from octonions by Inönü–Wigner contraction has a central generator, W, whose value fixes both the canonical position-momentum pairing and the strength of the non-associative structure. Is W genuinely central — in the representation-theoretic sense that would let it play the role of a fixed physical constant, the way $\hbar$ is fixed as the central element of the Heisenberg algebra — or only central in the bracket that defines the algebra itself?

Result. The algebra is a genuine Malcev algebra. Its universal nonassociative enveloping algebra therefore rigorously exists (Pérez-Izquierdo–Shestakov, 2004). W is centrally embedded in that envelope at **every** degree — proven by induction, not merely verified case by case, and independently confirmed computationally through degree 5. This gives W a structural role directly analogous to $\hbar$: fixed by Schur's lemma in any irreducible representation, and locked, by the algebra itself, to the strength of the non-associativity at the same linear order.

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1. The algebra

The constant R-flux algebra, obtained as an Inönü–Wigner contraction of the imaginary octonions under a $3+3+1$ generator split (Günaydin, Lüst & Malek, arXiv:1607.06474; independently reproduced and verified computationally in prior work), has seven generators $X^1, X^2, X^3, P_1, P_2, P_3, W$ and bracket structure

$$\{X^a, X^b\} = 2\epsilon^{abc}P_c, \qquad \{X^a, P_b\} = -2\delta^a_b\, W, \qquad \{P_a, P_b\} = 0,$$

$$\{X^a, W\} = 0, \qquad \{P_a, W\} = 0,$$

with W strictly central. Its Jacobiator is $J(X^a, X^b, X^c) = -12\epsilon^{abc}W$: nonzero, confirming genuine algebraic non-associativity. Fixing $W = -\frac{1}{2}$ recovers the canonical pairing $\{X^a, P_b\} = \delta^a_b$; more generally, $\{X^a, P_b\} = \kappa\delta^a_b$ requires $W = -\kappa/2$ and fixes the Jacobiator coefficient to exactly 6κ — the two quantities are not independent parameters of the algebra, but two faces of the single value W.

Everything below treats this bracket structure as given, established input; the octonionic derivation itself is not re-examined here.

2. The Malcev identity

A Malcev algebra is an anticommutative algebra whose bracket satisfies

$$J(x, y, [x, z]) = [J(x, y, z), x], \qquad J(x, y, z) := [[x, y], z] + [[y, z], x] + [[z, x], y].$$

This is strictly weaker than the Jacobi identity — Lie algebras are the special case $J \equiv 0$

— but it is a genuine, checkable constraint, not automatic for an arbitrary anticommutative bracket.

Verification. The identity was tested directly on the algebra of §1: across all $7^3 = 343$ basis triples, and separately across 100 random rational linear combinations of generators (to guard against identities that hold trivially on bare basis elements but fail on sums — a distinction that mattered elsewhere in this investigation and was checked for here as a matter of course). Zero violations were found in either test.

Worked example ($x = X^1, y = X^2, z = X^3$):

$$[x, z] = \{X^1, X^3\} = -2P_2, \qquad J(x, y, z) = J(X^1, X^2, X^3) = -12W.$$

$$J(x, y, [x, z]) = J(X^1, X^2, -2P_2) = 0, \qquad [J(x, y, z), x] = \{-12W, X^1\} = 0.$$

Both sides vanish, trivially matching here since W is central — the identity's content is exercised more fully on triples not involving W directly, all of which were checked with the same exhaustive procedure.

Why this is expected, not coincidental. The Malcev identity is a polynomial constraint on structure constants. For any $\lambda > 0$, the rescaled generators $X^a = \lambda x^a, P_a = \lambda^2 p_a$, $W = \lambda^3 w$ are related to the parent octonion generators by an invertible linear change of basis — the identity holds there trivially, by isomorphism invariance, since the parent algebra (imaginary octonions under commutator) is the canonical example of a non-Lie Malcev algebra. Continuity of the structure constants as $\lambda \to 0$ then carries the identity through the one genuinely singular step, where the transformation degenerates. The direct verification confirms this expectation rather than merely asserting it.

3. The Pérez-Izquierdo–Shestakov universal envelope

For any Malcev algebra M, Pérez-Izquierdo & Shestakov (*J. Algebra* 272, 2004) construct a universal nonassociative enveloping algebra

$$U(M) = F(M)/R(M),$$

where $F(M)$ is the free nonassociative algebra on a basis of M, and $R(M)$ is the two-sided ideal generated, for all $a, b \in M$ and $x, y \in F(M)$, by:

$$ab - ba - [a, b], \qquad (a, x, y) + (x, a, y), \qquad (x, a, y) + (x, y, a),$$

where $(p, q, r) := (pq)r - p(qr)$ is the associator. The first relation forces the commutator inside $U(M)$ to reproduce M's own bracket exactly — the direct nonassociative analogue of how an ordinary Lie algebra's enveloping algebra is defined.

A further theorem states that the canonical map $M \to U(M)$ lands inside the **generalized alternative nucleus**,

$$N_{\mathrm{alt}}(A) := \{a \in A : (a, x, y) = -(x, a, y) = (x, y, a) \text{ for all } x, y \in A\}.$$

This is a genuinely stronger statement than the two generating relations alone would obviously give: it holds for *arbitrary* $x, y \in U(M)$, of any degree, not merely for x, y drawn from M itself.

4. Centrality of W at degree 2

Derivation. For $x, y \in M$, expand $(xy)W$ and $W(xy)$ via the associator definition:

$$(xy)W = (x, y, W) + x(yW), \qquad W(xy) = (Wx)y - (W, x, y).$$

Since $W \in N_{\text{alt}}(U(M))$, the alternation property gives $(x, y, W) = (W, x, y)$. Using $Wx = xW$ (immediate from $[W, x] = 0$ and the first PIS relation):

$$[xy, W] = (xy)W - W(xy) = 2(W, x, y) + x(Wy) - (xW)y = 2(W, x, y) - (x, W, y).$$

Applying the alternation property once more, $-(x, W, y) = (W, x, y)$, gives the clean result

$$[xy, W] = 3(W, x, y), \qquad x, y \in M.$$

Evaluating the associator. The explicit formula for (a, b, x) with $a, b \in M, x \in U(M)$ of any degree (cf. Bremner et al., explicit degree-2 associator computations for low-dimensional Malcev algebra envelopes) is

$$(a, b, x) = \tfrac{1}{6}[[x, a], b] - \tfrac{1}{6}[[x, b], a] - \tfrac{1}{6}[x, [a, b]].$$

Worked example ($a = W,\ b = X^1,\ x = X^2$):

$$[x, a] = \{X^2, W\} = 0 \implies [[x, a], b] = 0,$$

$$[x, b] = \{X^2, X^1\} = -2P_3 \implies [[x, b], a] = \{-2P_3, W\} = 0,$$

$$[a, b] = \{W, X^1\} = 0 \implies [x, [a, b]] = 0.$$

$$\therefore\ (W, X^1, X^2) = \tfrac{1}{6}(0) - \tfrac{1}{6}(0) - \tfrac{1}{6}(0) = 0.$$

This vanishing was checked exhaustively: $(W, X_i, X_j) = 0$ for all six ordered pairs $i \neq j$, and likewise for all (W, P_i, P_j) and (W, X_i, P_j) — every degree-2 combination of the six non-central generators.

Mechanism. The vanishing is not automatic from W's centrality alone — a generic central element need not survive this construction. It happens here because two independent centrality facts combine: W commutes with the X's directly (killing the first and third terms above), and the one surviving term, $[[X_i, X_j], W]$, involves $[X_i, X_j] \propto P_k$ — which is *also* annihilated by W, since $\{P, W\} = 0$ as well. W is central with respect to both halves of the algebra simultaneously, and the one term that could have broken centrality is caught by that same double centrality one step removed.

$$\therefore \quad [xy, W] = 0 \text{ for every degree-2 product of the algebra's generators.}$$

5. Centrality of W at every degree

The general reduction lemma. For a basis monomial x of $U(M)$ with degree ≥ 2, write $x = by$ with $b \in M$ (the left-tapped monomial basis makes this decomposition canonical). Then, for any $a \in M$ (Pérez-Izquierdo–Shestakov; Bremner, Hentzel, Peresi, Tvalavadze & Usefi, arXiv:1008.2388, Lemma 1):

$$[x, a] = [by, a] = [b, a]y + b[y, a] + \tfrac{1}{2}[[y, a], b] - \tfrac{1}{2}[[y, b], a] - \tfrac{1}{2}[y, [a, b]].$$

This is stated with no restriction on y's own degree — it is a genuinely general, degree-independent reduction, not a formula specific to $|x| = 2$ or $|x| = 3$. Crucially, every term on the right involves either y directly or a bracket applied to y — strictly lower degree than x itself.

Theorem. W is centrally embedded in $U(M)$ at every degree: $[x, W] = 0$ for every basis monomial x of $U(M)$.

Proof, by induction on degree. *Base case* (deg $x = 1$): $x \in M$, and $[x, W] = 0$ by W's centrality in M itself.

Inductive step. Assume $[n, W] = 0$ for every monomial n of degree $\leq d - 1$. Let $x = a \cdot n$ have degree d, $a \in M$ a generator. Expanding via the associator definition and the N_{alt} alternation property (§3, valid for arbitrary elements of $U(M)$, any degree) exactly as in §4's degree-2 derivation:

$$[an, W] = 2(W, a, n) + a(Wn) - (aW)n.$$

By the inductive hypothesis, $Wn = nW$; substituting and applying alternation once more gives the same clean reduction found at degree 2 and degree 3, now shown to hold generally:

$$[an, W] = 3(W, a, n).$$

Evaluating the associator via the formula of §4, $(W, a, n) = \frac{1}{6}[[n, W], a] - \frac{1}{6}[[n, a], W] - \frac{1}{6}[n, [W, a]]$: the first term vanishes by the inductive hypothesis ($[n, W] = 0$), the third vanishes since $[W, a] = 0$ (W central in M). So

$$(W, a, n) = -\tfrac{1}{6}[[n, a], W].$$

By the general reduction lemma above, $[n, a]$ is a sum of terms each involving y (or a bracket applied to y), where $n = by$ — every such term has degree strictly less than n's own degree, hence degree $\leq d - 2 < d - 1$. By the inductive hypothesis, W commutes with every one of them. Hence $[[n, a], W] = 0$, so $(W, a, n) = 0$, so $[an, W] = 0$. ■

This is a complete proof, not an extrapolation: the inductive step uses only the alternation property (general, any degree) and the reduction lemma (general, any degree), closing at every step without appeal to anything specific to a particular degree.

Computational confirmation. The full recursive reduction — both the commutator lemma above and its companion for left-multiplication $a \cdot x$ (Bremner et al., Lemma 2), needed to evaluate the nested terms concretely — was implemented directly and used to verify $[x, W] = 0$ exhaustively at degrees 2, 3, 4, and 5: all $6^2 = 36, 6^3 = 216, 6^4 = 1296$, and $6^5 = 7776$ generator combinations respectively, zero violations at any degree. This is independent confirmation that the inductive argument's logic holds correctly at each concrete step, not merely in the abstract.

Worked illustration (degree 3, $x = X^1, y = X^2, z = X^3$): the degree-1 remainder in $[yz, x]$'s expansion vanishes directly by W's centrality in M; the degree-2 pieces decompose into generator components each individually confirmed by §4. Combining gives $[x(yz), W] = 0$, matching the general theorem.

Mechanism, restated in general form. The reduction at every degree traces back to the same double-centrality identified at degree 2 (§4): W commutes with every generator directly, and everything those generators' brackets produce is, by the inductive hypothesis, also already known to commute with W. The induction is really just this mechanism, applied once per degree.

$$\therefore \quad [x, W] = 0 \text{ for every basis monomial of } U(M), \text{ at every degree.}$$

6. Physical interpretation

A central element of a universal enveloping algebra is forced, by Schur's lemma, to act as a fixed scalar multiple of the identity in any irreducible representation — the precise structural reason $\hbar$ takes one universal value as the central element of the Heisenberg algebra, rather than varying representation to representation. §5 establishes that W occupies exactly this role for the R-flux algebra's representation theory, proven at every degree rather than assumed by resemblance or left as a bounded conjecture.

Combined with the locked relation from §1 — $\{X^a, P_b\} = \kappa\delta^a_b \iff W = -\kappa/2 \iff$ Jacobiator coefficient $= 6\kappa$ — this is not merely a fact about the algebra's defining bracket, but a property inherited into its actual representation theory: a genuine quantization built on this structure carries a single, fixed "quantum of non-associativity," arithmetically tied to $\hbar$ by the algebra itself, rather than these being freely independent parameters as in generic non-geometric flux phenomenology (where the R-flux scale, typically $\propto l_p^3$, is treated as tunable independently of $\hbar$).

In the classical limit $W \to 0$: since both the canonical pairing's scale and the Jacobiator's coefficient are linear in W at fixed ratio, quantum uncertainty and algebraic non-associativity vanish at *exactly* the same rate — a specific, structurally forced, falsifiable-in-principle claim, and a genuine departure from treating them as independently tunable.

7. Open questions

- **Uniqueness.** §5's result holds for the specific Günaydin–Lüst–Malek contraction of the imaginary octonions. Whether an analogous central, $\hbar$-like element and locked ratio arise from other constructions of algebraic R-flux is untested.

- **Beyond centrality.** W being central establishes that it acts as a scalar in any irreducible representation; it does not by itself determine the fuller representation theory of $U(M)$ — its explicit structure constants, or the classification of its irreducible representations — which remains substantial further work, following the low-dimensional cases already worked out by Bremner et al. for other Malcev algebras.

A related question is answered in a companion paper rather than left open here: how canonical quantum mechanics — associative, with a nonzero central element — relates to this structure at all, given that the Jacobi identity forces $W = 0$ in any associative bracket representation of the full algebra. The resolution (*The Second Contraction*) is a further Inönü–Wigner contraction, with weights $(1, 1, 2)$, under which the Jacobiator dies linearly while the pairing $\{X, P\} \sim W$ is preserved exactly, landing on the $3+3+1$ Heisenberg algebra with the central element alive.

References

- Companion paper: *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2.*
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